

OPERATING SYSTEMS

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CO₂ - AT1

CSA0701

(Analytical problem solving)

Q.1.

Process

Burstfimo

PI

10

amire - 0

P2

4

P 3

6

PH

2

a) Calc waiting time & Turnaround time using (SJF) Shortest Job first

b) Compute the avg waiting & turnaround time

c) Interpret whether SJF improves CPU utilization compared to FCFS for the workload given.

a) Now: Turnaround time $\Rightarrow$ Arrival time

waiting time $\Rightarrow$ Turnaround - Bursttime

Proc	Burst time	Turnaround time	Waiting time
P1	10	22	12
P2	4	6	2
P3	6	12	6
P4	2	2	0

P_4	P_2	P_3	P_1
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2

6 +

12

22

b) i) Avg waiting time =

$$\frac{12 + 2 + 6 + 0}{4} = 5 \text{ ms}$$

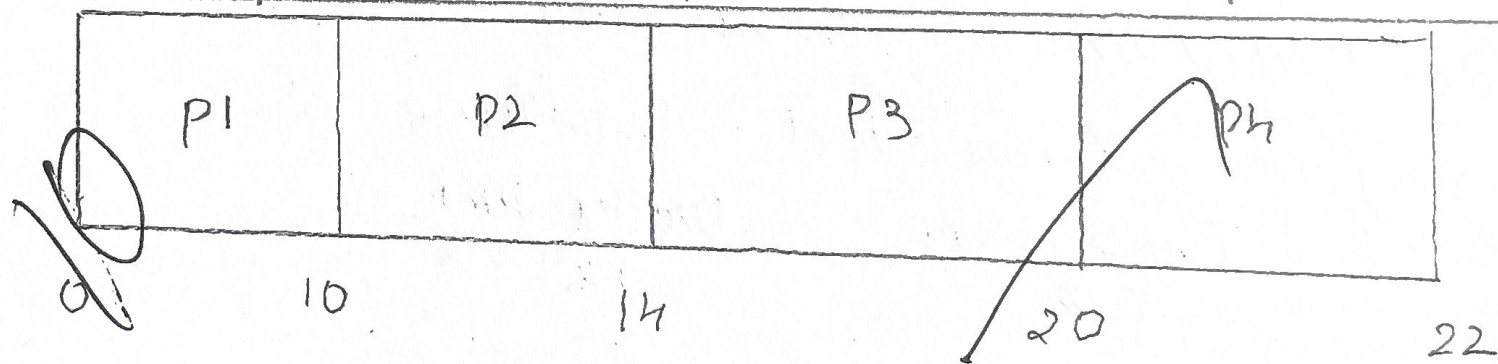
ii) Avg turnaround time

$$\frac{22 + 6 + 12 + 2}{4} = 10.5 \text{ ms}$$

Round Robin
priority
P1

c) Now FCFS

Process	Burst time	Turnaround CT - AT	Waiting time TT - BT
P1	10	10	0
P2	4	14	10
P3	6	20	14
P4	2	22	20



Now:

Avg waiting time: $\frac{0 + 10 + 14 + 20}{4} = 11 \text{ ms}$

Avg ~~burst~~ turnaround time: $\frac{10 + 14 + 20 + 22}{4} = 16.5 \text{ ms}$

On comparing SJF with FCFS, The average waiting time of (Shortest Job first) is less compared to FCFS, Thus SJF is considered best & fastest.

Round Robin Scheduling

Quantum time = 4 ms

Process	Burst time
P1	20
P2	5
P3	13
P4	8

- Construct the Gantt chart for Round Robin
- Calculate ^{avg} waiting time & Turnaround time of each process.
- Analyze how changing $q = 10$ ms affects CPU performance utilization & context switching
- Deadlock - Safety check using Banker's algorithm.

A system has 3 resources: $A=10, B=5, C=7$

Allocation Matrix

Process.

P1

P2

P3

P4

P5

A

B

C

0

1

0

0

0

2

0

2

3

1

1

2

0

2

0

Max Matrix

process

P1

P2

P3

P4

P5

A

B

C

7

5

3

3

2

2

0

0

2

4

2

2

5

3

3

73.79.1

a) Calculate the Need Matrix

b) Determine if the system is in a safe state using Banker's Algorithm.

Avail - Total

Need

Algorithm

c) Provide a valid & safe sequence if it exist.

2. Ans:

Process	Burst time	Complete time		TT
P1	20	42	Req = 4ms	46
P2	5 ✓	21		21
P3	13 ✓	46		42
P4	8 ✓	29		29

P1	P2	P3	P4	P1	P2	P3
0	4	8	12	16	20	21
						25

P4	P1	P3	P1	P3	P1
25	29	33	37	41	42
					46

b) Avg WT = $(26 + 16 + 28 + 21) / 4 = 23 \text{ ms}$

Avg TAT = $(46 + 21 + 42 + 29) / 4 = 34.5 \text{ ms}$

c) $\Rightarrow$ CPU utilization increases much as only less time is spent between the switching of process
 $\Rightarrow$ Round robin behaves more like FCFS (scheduling)
 $\Rightarrow$ Users may wait longer before the process get turned

Need Matrix $\Rightarrow$ Max-Allocation

P	A	B	C
P1	7	4	3
P2	1	2	2
P3	6	0	0
P4	2	1	1
P5	5	3	1

Allocation $\rightarrow$ given

Available = Total - Allocation

$$A = 10 - 7 = 3$$

$$B = 5 - 2 = 3$$

$$C = 7 - 5 = 2$$

Available (3, 3, 2)

Allocated

$$A = 0 + 2 + 3 + 2 + 0 = 7$$

$$B = 1 + 0 + 0 + 1 + 0 = 2$$

$$C = 0 + 0 + 2 + 1 + 2 = 5$$

New available 2

Now:

$P_2 \rightarrow$ Need (1, 2, 2) $\rightarrow$ Available (3, 3, 2) $\Rightarrow$ (7, 4, 3)

$P_4 \rightarrow$ Need (2, 1, 1) $\rightarrow$ Available (7, 4, 3) $\Rightarrow$ (7, 4, 5)

$P_5 \rightarrow$ Need (3, 3, 1) $\rightarrow$ Available (7, 4, 5) $\Rightarrow$ (10, 5, 5)

$P_1 \rightarrow$ Need (7, 4, 3) $\rightarrow$ Available (10, 5, 5) $\Rightarrow$ (10, 5, 7)

$P_3 \rightarrow$ Need (6, 0, 0) $\rightarrow$ Available (10, 5, 7) $\Rightarrow$ (10, 5, 7)

Thus it is in safe state

Thus safe sequence is

$P_2 \rightarrow P_4 \rightarrow P_5 \rightarrow P_1 \rightarrow P_3$

c)

A system has the following resource allocation

- P_1 is holding R_1 and waiting for R_2
- P_2 is holding R_2 and waiting for R_3
- P_3 is holding R_3 and waiting for R_1

a) Draw resource allocation graph

b) Determine whether system is in deadlock

c) Recommend a strategy to resolve it & justify the method.

Soln:

Allocation edges:

$R_1 \rightarrow P_1$

$R_2 \rightarrow P_2$

$R_3 \rightarrow P_3$

Request edges

$P_1 \rightarrow R_2$

$P_2 \rightarrow R_3$

$P_3 \rightarrow R_1$

b) System is a deadlock $\rightarrow$ Yes

cycle:

$P_1 \rightarrow R_2 \rightarrow P_2 \rightarrow R_3 \rightarrow P_3 \rightarrow R_1 \rightarrow P_1$

Since each process is waiting for a resource held by another process, the system is in a deadlock.

c) Recovery method:

$\Rightarrow$ Terminating one process (Eg: P_3)

$\Rightarrow$ Force one process to release its resource. This breaks the cycle and allows the other processes to continue.

5 A system has the following process behavior:
• Each process spends 20% time in CPU & 80% waiting for I/O

• Degree of multi programming = 4 processes.
CPU utilization formula:

$$U = 1 - (ph)^n \text{ when,}$$

p = prob of process h = no of process in memory

- Calculate CPU utilization.
- Compute CPU utilization if the degree of multiprogramming increases to 6
- Interpret which configuration improves system performance & why.

Soln:

a) Degree of Multi programming = 4

$$U = 1 - (0.8)^4$$

$$U = 1 - 0.4096 = 0.5904$$

$$\text{CPU utilization} = 59.04\%$$

b) Degree of Multiprogramming = 6

$$U = 1 - (0.8)^6$$

$$U = 1 - 0.262144 = 0.737856$$

$$\text{CPU utilization} = 73.79\%$$

c) Interpretation:

4 processes : CPU utilization : 59.04%

6 processes : CPU utilization : 73.79%